

NIELIT O LEVEL • UNIT 08

Web Publishing and Browsing

Curriculum Hours: 5h (2h Theory + 3h Lab) | Topics: 13 | Language: English

UNIT OVERVIEW & PEDAGOGICAL SCOPE

Web Publishing and Browsing represents the final capstone phase of web engineering: taking locally developed web documents and making them globally accessible to billions of internet users. This unit teaches the end-to-end Web Publishing lifecycle, user-centric web design principles, the SGML heritage of markup languages, web hosting architectures (Shared, VPS, Dedicated, Cloud), international document interchange standards (ASCII, UTF-8, MIME, XML), essential server-side publishing components (Domain registrars, DNS records, Apache/Nginx web servers), site document directory management, search engine architectures (crawlers, indexing, PageRank, SEO), meta search engine aggregators (Dogpile), WWW architecture and URL anatomy, web browser internal architecture and rendering engines (Blink, Gecko, WebKit), the stateless HTTP protocol (methods GET/POST and status code ranges 200/404/500), and modern publishing tools including FileZilla FTP and cPanel.

WHAT YOU WILL MASTER IN THIS UNIT

- The complete 7-stage Web Publishing Lifecycle: Planning, UI/UX Design, Development, Testing, Domain & Hosting, Launch, and Maintenance.
- Core Web Design Principles: Visual hierarchy, intuitive navigation, F-shaped scanning pattern, mobile responsiveness, and page load optimization.
- SGML (Standard Generalized Markup Language ISO 8879:1986): Relationship to HTML and XML.
- Web Hosting models compared: Shared Hosting, VPS (Virtual Private Server), Dedicated Server, and Cloud Hosting.
- Document Interchange Standards: 7-bit ASCII, universal UTF-8 Unicode, MIME types, and XML data interchange.
- Components of Web Publishing: Domain Registrars, DNS name servers, Web Servers (Apache, Nginx, IIS), and Database Servers.
- Document Management: `public\_html` root directory conventions, absolute versus relative linking, and clean naming rules.
- Search Engine mechanics: Automated Web Crawlers/Spiders, indexing databases, ranking algorithms, and On-Page/Off-Page SEO.
- Meta Search Engines: How aggregators (like Dogpile) query multiple search engines simultaneously.
- WWW Architecture & URL Anatomy: Scheme protocol, subdomain, domain name, port numbers (80/443), path, query strings, and hash fragments.
- Browser Architecture: The 7-layer browser internal model and rendering engines (Blink in Chrome/Edge, Gecko in Firefox, WebKit in Safari).
- HTTP / HTTPS Protocol: Stateless communication, Request-Response cycle, HTTP Methods (GET, POST, PUT, DELETE), and Status Codes (200 OK, 301, 403, 404, 500).
- Publishing Tools: Deploying websites using FileZilla FTP/SFTP client over Port 21/22 and cPanel hosting administration.

TOPIC 8.1: OVERVIEW OF WEB PUBLISHING

Web Publishing (also called Online Publishing) is the complete end-to-end process of designing, developing, testing, uploading, and maintaining a website on an Internet-connected web server so it is publicly accessible via a registered domain name.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Writing HTML/CSS files on a local laptop is useless until they are published to a public web server where users worldwide can access them 24 hours a day, 365 days a year.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Web publishing follows a disciplined 7-stage engineering lifecycle from initial client concept to post-launch maintenance.

THE 7 PHASES OF THE WEB PUBLISHING LIFECYCLE

- 1. Project Planning & Strategy: Defining target audience, site goals, sitemap architecture, and required technical stack.
- 2. Wireframing & UI/UX Design: Creating visual mockups, color palettes, and interactive prototypes in tools like Figma or Photoshop.
- 3. Front-End & Back-End Development: Coding structured HTML, CSS, JavaScript, server APIs, and relational databases.
- 4. Quality Assurance & Testing: Cross-browser testing, mobile viewport verification, broken link checks, and accessibility audits.
- 5. Domain Registration & Web Hosting: Purchasing a unique domain name (`.com`, `.in`) and leasing server space with an ISP/Host.
- 6. Deployment (Publishing): Uploading code and assets to the server's `public\_html` directory using FTP/SFTP or Git.
- 7. Maintenance & SEO Optimization: Monitoring uptime, renewing SSL certificates, fixing bugs, updating content, and improving search engine ranking.

A local college wants a web presence: (1) Plan pages: Home, Courses, Admission. (2) Design layout in Photoshop. (3) Code HTML/CSS/JS. (4) Test on mobile and laptop. (5) Register `mycollege.edu.in`. (6) Upload files using FileZilla to server. (7) Website is live globally!

- Web Publishing encompasses the entire lifecycle from planning to continuous maintenance.
- Websites are published by uploading files to a public web server.
- Testing across different browsers and mobile devices is mandatory before public launch.

QUICK TOPIC RECAP

Web publishing is the process of creating, deploying, and maintaining websites on public web servers across 7 phases: Planning -> Design -> Development -> Testing -> Hosting -> Deployment -> Maintenance.

TOPIC 8.2: WEB PAGE DESIGN CONSIDERATIONS AND PRINCIPLES

Web design principles are established usability guidelines, visual hierarchies, and cognitive ergonomics that dictate how information should be organized on a webpage to maximize readability, accessibility, and user retention.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Visitors decide within 3 seconds whether to stay on a website or bounce away. Poor color contrast, cluttered navigation, or slow loading instantly drives users to competitors.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Designers employ visual psychology, color theory, standardized typography scales, and eye-tracking patterns to guide user attention smoothly down the page.

CORE WEB DESIGN PRINCIPLES REFERENCE

Design Principle	Core Concept / Rule	Why It Matters & User Impact
Simplicity (KISS Principle)	'Keep It Simple, Stupid': Eliminate unnecessary visual clutter, excessive popups, and gratuitous animations	Keeps users focused on core content and primary calls-to-action (CTA)
Visual Hierarchy	Arrange elements in order of visual importance using font size, color weight, and spacing	Guides reader's eyes naturally from main H1 headlines down to supporting body text
F-Shaped Scanning Pattern	Web users scan screens in an 'F' shape: reading across top, across middle, then scanning left edge down	Crucial for placing logos, headlines, and key navigational links where eyes naturally look first

Mobile Responsiveness	Layout must adapt gracefully to smartphones, tablets, and wide desktop screens	Ensures high mobile conversion rates and prevents mobile bounce penalties from Google
Page Load Speed	Optimize images, minify code, and leverage browser caching to load pages under 2 seconds	Every 1-second delay in page load time reduces visitor satisfaction by 16%
Accessible Contrast	Maintain high contrast between text and background (e.g. dark text on light background)	Ensures readability for elderly users, people in bright sunlight, and visually impaired users

Applying the F-Pattern: Place the university logo in the top-left corner (first glance). Place the main navigation bar across the top horizontal band. Place high-yield news headlines down the left column where eye tracking is strongest.

- Web users scan content in an F-shaped reading pattern rather than reading every word.
- Visual hierarchy uses size, contrast, and spacing to indicate importance.
- Websites should ideally load in under 2 seconds to retain visitors.
- High contrast between text and background is mandatory for web accessibility.

QUICK TOPIC RECAP

Design principles: Simplicity (declutter), Visual Hierarchy (size indicates priority), F-Pattern (place key content along top and left), Mobile Responsiveness, and Fast Load Speed.

TOPIC 8.3: SGML (STANDARD GENERALIZED MARKUP LANGUAGE)

SGML (Standard Generalized Markup Language, ISO 8879:1986) is an international standard metalanguage used for defining structured generalized markup languages for electronic documents.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

SGML is the historical parent of both **HTML** and **XML**. Understanding SGML clarifies why HTML uses tags, doctypes, and document type definitions (DTDs).

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

SGML provides a formalized grammar to create markup languages. A Document Type Definition (DTD) specifies which tags exist, which attributes they accept, and how they can be nested.

THE FAMILY TREE: SGML, HTML, AND XML

- SGML (ISO 8879, 1986): The complex parent metalanguage. Powerful, highly generalized, but too large and complicated to run natively inside web browsers.
- HTML (Application of SGML): Sir Tim Berners-Lee took a small, specialized subset of SGML tags specifically optimized for hyperlinked academic and web documents (HTML 2.0 to HTML 4.01 were formal SGML applications).
- XML (Extensible Markup Language): Created in 1998 by W3C as a simplified, strict, lightweight subset of SGML designed for structured data interchange across the Internet.
- HTML5 Independence: HTML5 formally broke away from SGML restrictions, abandoning strict DTD requirements in favor of a modern DOM-based Living Standard (`<!DOCTYPE html>`).

In HTML 4.01, the doctype was a massive SGML reference: `<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">`. In HTML5, because it is no longer bound to SGML DTDs, it simplified to just `<!DOCTYPE html>`.

- SGML stands for Standard Generalized Markup Language (ISO 8879:1986).
- HTML (up to version 4.01) is a specialized application of SGML.
- XML is a simplified, lightweight subset of SGML designed for data exchange.
- HTML5 is no longer based on SGML DTDs.

QUICK TOPIC RECAP

SGML is the parent metalanguage of markup. HTML is a specialized application of SGML for web documents; XML is a simplified subset of SGML for data interchange.

TOPIC 8.4: WEB HOSTING BASICS (SHARED, VPS, DEDICATED, CLOUD)

Web Hosting is an internet service that leases physical server storage space, network bandwidth, and compute power to individuals and organizations, allowing their websites to be accessed 24x7 over the Internet.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

A website cannot be viewed online unless its files reside on a permanently powered, high-speed Internet-connected computer (a Web Server).

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Web hosting companies (e.g. Hostinger, AWS, Bluehost, GoDaddy) operate massive data centers containing racks of server blades with redundant power (UPS) and gigabit fiber connections.

THE 4 MAJOR WEB HOSTING MODELS COMPARED

Hosting Category	Physical Architecture / Isolation	Pros & Performance	Cons & Typical Use Case
Shared Hosting	Hundreds of websites share the exact same physical CPU, RAM, and storage	Extremely cheap (~ 100/mo); easy cPanel admin; zero technical setup needed	Slow; 'bad neighbor effect' (one site's traffic surge slows all other sites); beginner blogs
VPS (Virtual Private Server)	One physical server partitioned into isolated virtual servers using a Hypervisor	Guaranteed dedicated CPU/RAM slices; root access; higher reliability and security	Moderate cost; requires some server administration knowledge; medium-traffic e-commerce
Dedicated Server	An entire physical computer leased exclusively to a single client organization	100% maximum performance, total hardware control, enterprise-grade security	Very expensive; requires full-time system administration team; banks, universities, portals
Cloud Hosting	Website runs across a clustered network of interconnected virtual servers (AWS, Azure)	**Elastic scalability** (scales resources automatically); pay-as-you-go; 99.99% uptime	Variable monthly billing costs; modern high-traffic startups, streaming, SaaS platforms

A student creating an O-Level project uses ****Shared Hosting**** (costs 99/month, simple). A growing startup getting 50,000 visitors uses ****VPS****. An airline booking system or university portal uses ****Cloud Hosting**** to scale up 50 servers during festival sales.

- Shared hosting shares CPU, RAM, and disk with hundreds of other websites.
- VPS (Virtual Private Server) uses virtualization to give dedicated slices of resources.
- Dedicated hosting provides an entire physical server dedicated to one customer.
- Cloud hosting distributes website load across a cluster of interconnected virtual servers.

QUICK TOPIC RECAP

Hosting models: Shared (cheapest, shared resources), VPS (virtual isolation), Dedicated (full physical server for one client), Cloud (elastic clustering, pay-as-you-go).

TOPIC 8.5: DOCUMENTS INTERCHANGE STANDARDS (ASCII, UTF-8, MIME, XML)

Document interchange standards are universally agreed-upon technical formats and character encodings that enable heterogeneous computer operating systems (Windows, Linux, macOS, Android) to exchange data and documents without corrupting characters or file structures.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Without standardized character sets and MIME headers, an English text file sent from a Linux server to a Windows browser would display unreadable garbled characters (Mojibake).

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Standard protocols specify character mapping tables (ASCII, Unicode) and content type declarations (MIME) that all browsers decode identically.

INTERCHANGE STANDARDS SPECIFICATION

Standard / Acronym	Full Name & Architecture	Key Technical Specification	Primary Role in Web Computing
ASCII	American Standard Code for Information Interchange (1963)	7-bit encoding representing exactly 128 characters (0-127: English letters, numbers, symbols)	Foundational standard; limited strictly to basic English characters
UTF-8	Unicode Transformation Format (8-bit variable width)	Variable length (1 to 4 bytes per character); encodes all 1,112,064 Unicode code points	Universal Web Standard : Supports English, Hindi, Arabic, Chinese, emojis, math symbols
MIME	Multipurpose Internet Mail Extensions	Header standard categorizing digital media types ('Content-Type: text/html')	Informs browsers whether an incoming file is HTML, a JPEG image, a PDF, or a video
XML	Extensible Markup Language (W3C Standard)	Text-based structured markup format with user-defined semantic tags ('<student><name>...')	Standard data exchange format used in web services, RSS feeds, and configuration files

When an Indian e-commerce receipt displays ₹ (Indian Rupee symbol) and Hindi text 'नमस्ते': **UTF-8** ensures both characters render accurately. The server transmits 'Content-Type: text/html; charset=UTF-8' (using **MIME**), instructing the browser how to decode the bytes.

- ASCII is a 7-bit character encoding standard supporting 128 characters.
- UTF-8 is the universal variable-width encoding used across more than 98% of all websites.
- MIME types identify the file format sent over HTTP (e.g. 'text/html', 'image/png', 'application/pdf').
- XML is a hardware- and software-independent markup language designed to transport and store data.

QUICK TOPIC RECAP

Interchange standards: ASCII (7-bit, 128 English chars), UTF-8 (universal variable 1-4 bytes for all world languages/emojis), MIME (file format headers), XML (data transport).

TOPIC 8.6: COMPONENTS OF WEB PUBLISHING

Components of Web Publishing refer to the complete technical ecosystem of hardware, software, network protocols, and third-party services required to take web files from a local computer and publish them online.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

A web developer must understand every component in the publishing pipeline to properly configure domains, connect DNS name servers, manage databases, and troubleshoot deployment failures.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

The publishing infrastructure connects five distinct architectural components: Registrars, DNS Servers, Web Server Software, FTP/SSH Services, and Database Servers.

THE 5 CORE HARDWARE & SOFTWARE PUBLISHING COMPONENTS

- 1. Domain Name Registrar: ICANN-accredited commercial agencies (e.g. GoDaddy, Namecheap, BigRock) where users register and renew human-readable domain names (e.g. `nielit.gov.in`).
- 2. DNS Name Servers: Specialized server computers storing DNS zone files that map registered domain names to the hosting server's numeric IP address (`A Records`).
- 3. Web Server Software: Server-side application software running on port 80/443 that accepts incoming HTTP requests and serves web documents (e.g. **Apache HTTP Server**, **Nginx**, **Microsoft IIS**).
- 4. FTP / SFTP Server: File Transfer Protocol service (running on port 21/22) allowing developers to securely upload files from their local laptop directly into the server's storage.
- 5. Database Server: Dedicated Relational Database Management System (RDBMS) software (e.g. **MySQL**, **MariaDB**, **PostgreSQL**) storing dynamic customer accounts, articles, and product catalogs.

You buy the domain `myschool.org` from a Registrar. You point its DNS Name Servers to your hosting account. You launch FileZilla (FTP) and upload `index.html`. Your web server (Apache) listens on Port 80 and serves your page to the world.

- ICANN (Internet Corporation for Assigned Names and Numbers) oversees global domain registration.
- Popular web server software includes Apache, Nginx, and Microsoft IIS.
- FTP (File Transfer Protocol, Port 21) or SFTP (Port 22) is used to upload files to web servers.
- MySQL is the most popular open-source database server used in web publishing.

QUICK TOPIC RECAP

Web publishing requires 5 components: Domain Registrar (buys name), DNS (maps name to IP), Web Server (Apache/Nginx serves HTML), FTP/SFTP (uploads files), and Database (MySQL).

TOPIC 8.7: DOCUMENT MANAGEMENT & SITE ARCHITECTURE

Document Management is the systematic organization of web project files, asset folders, directory paths, and file permissions on a web server's storage drive.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Unorganized files with broken links, incorrect relative paths, or wrong server permissions cause 404 Not Found errors, broken styling, and security vulnerabilities.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Web servers designate a specific root folder-conventionally named `public_html`, `www`, or `htdocs`-as the public-facing directory. Any file placed inside this folder is visible to the public internet.

ABSOLUTE VS. RELATIVE FILE PATHING

- Relative Path (Best Practice): Points to files relative to current folder (e.g. `images/logo.png` or `../css/style.css`). Highly portable: works identically on local laptop and live server.
- Absolute Path: Specifies full URL (e.g. `https://nielit.gov.in/images/logo.png`). Breaks if domain name changes.
- Index Page Rule: Web servers automatically look for `index.html` (or `index.php`) when a user visits a directory root without specifying a file name.
- File Permissions (Linux/Unix): Web files should have permission `644` (read/write for owner, read-only for public); directories should have permission `755`.

STANDARD WEB DIRECTORY STRUCTURE HIERARCHY

Directory / Path	Folder Purpose	Standard Content Stored Inside
<code>/public_html/</code> (Root)	The primary public root directory of the website	<code>index.html</code> (homepage), <code>robots.txt</code> , <code>.htaccess</code>
<code>/public_html/css/</code>	Dedicated subfolder for Cascading Style Sheets	<code>style.css</code> , <code>w3.css</code> , <code>responsive.css</code>
<code>/public_html/js/</code>	Dedicated subfolder for JavaScript client scripts	<code>app.js</code> , <code>validation.js</code> , <code>angular.min.js</code>
<code>/public_html/images/</code>	Dedicated subfolder for graphics and photographs	<code>logo.svg</code> , <code>banner.jpg</code> , <code>icons/</code>
<code>/public_html/docs/</code>	Dedicated subfolder for downloadable public assets	<code>syllabus.pdf</code> , <code>admit-card-instructions.pdf</code>

If `contact.html` needs to display an image stored in the `images` folder: Write ``. The relative path tells the browser to look inside the sibling `images` directory.

- The public root directory on standard web servers is named `public_html` or `htdocs`.
- Relative file paths (e.g. `css/style.css`) are best practice for portable web development.
- Standard Linux server file permissions are 644 for files and 755 for directories.

QUICK TOPIC RECAP

Organize files in `public_html`: store CSS in `css/`, scripts in `js/`, images in `images/`. Always use relative paths (`css/style.css`) and name the homepage `index.html`.

TOPIC 8.8: SEARCH ENGINES (CRAWLERS, INDEXING, RANKING, SEO)

A Search Engine is a specialized online software system (such as Google, Microsoft Bing, or Yahoo) that continuously discovers, crawls, indexes, and ranks billions of web pages across the World Wide Web so users can find information using keyword queries.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Publishing a website is pointless if nobody can find it. Over 70% of all website traffic originates from search engines. Search Engine Optimization (SEO) ensures high search visibility.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Search engines operate through three continuous automated stages: 1. **Crawling** (discovering pages), 2. **Indexing** (storing and cataloging content in massive databases), and 3. **Ranking** (sorting results by relevance when a user searches).

THE 3 PILLARS OF SEARCH ENGINE ARCHITECTURE

- 1. Web Crawlers (Spiders / Bots): Automated software robots (e.g. Googlebot) that continuously traverse the Internet by following hyperlinks from one webpage to another, downloading page content.
- 2. The Index: A colossal, distributed database containing trillions of cataloged web pages, inverted word lists, and metadata.
- 3. Ranking Algorithms (e.g. PageRank): Complex mathematical models evaluating over 200 ranking signals (content relevance, backlinks, mobile speed, SSL security) to display the most helpful results on the Search Engine Results Page (SERP).
- robots.txt: A plain text file placed in the website root instructing crawlers which directories they are permitted or forbidden to scan.
- sitemap.xml: An XML file listing all public page URLs on a website to help search crawlers discover new content quickly.

ON-PAGE SEO VS. OFF-PAGE SEO

SEO Category	Core Focus & Responsibility	Practical Techniques Applied by Web Designers
On-Page SEO	Optimizations made directly on the website's HTML code and content	Descriptive ` <code><title></code> ` tags, meta descriptions, semantic ` <code><h1></code> ` to ` <code><h6></code> ` hierarchy, ` <code>alt`</code> text on images, clean URLs, mobile responsive design, fast page load speeds
Off-Page SEO	Actions taken outside the website to build external authority and trust	Acquiring high-quality external backlinks from authoritative websites, social media shares, institutional citations, brand mentions

To rank an O-Level course page on Google: Add ``<title>NIELIT O-Level M2-R5.1 Notes - Web Designing</title>``, use semantic ``<h1>`` tags, add ``alt="Syllabus diagram"`` to images, and create a ``sitemap.xml``. Googlebot crawls the page and ranks it at the top of search results.

- Search engines operate in 3 stages: Crawling (discovery), Indexing (storage), and Ranking (ordering).
- Automated software programs that discover web pages are called Web Crawlers, Spiders, or Bots.
- SEO stands for Search Engine Optimization.
- ``robots.txt`` controls crawler access; ``sitemap.xml`` lists pages for fast crawler discovery.

QUICK TOPIC RECAP

Search engines discover pages via Crawlers (spiders), catalog them in an Index, and order results using Ranking algorithms. On-page SEO optimizes HTML tags, titles, and speed.

TOPIC 8.9: META SEARCH ENGINES

A Meta Search Engine is an information retrieval aggregator that does not maintain its own database of web pages or web crawlers; instead, it simultaneously queries multiple independent search engines (such as Google, Bing, and Yahoo), aggregates their results, removes duplicates, and presents a consolidated results list to the user.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

No single search engine indexes 100% of the World Wide Web. Meta search engines provide broader coverage, comparative perspectives, and eliminate single-engine bias.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

When a user types a query, the meta search engine contacts multiple search engine APIs simultaneously, receives raw result feeds, strips duplicate URLs, computes an aggregated relevance score, and displays the unified list.

SEARCH ENGINES VS. META SEARCH ENGINES

Attribute / Parameter	Standard Search Engine (e.g. Google, Bing)	Meta Search Engine (e.g. Dogpile, Metacrawler)
Underlying Database	Maintains its own massive multi-petabyte database index	**Has NO database of its own**; queries external engines
Web Crawlers / Spiders	Builds and operates fleets of automated web crawlers	**Has NO web crawlers**; relies entirely on third-party crawlers

Result Generation	Queries its internal database index directly	Queries multiple external search engines simultaneously via API
Coverage Scope	Limited strictly to what its own crawlers have discovered	Combines coverage from multiple search engine indexes
Prominent Examples	Google, Microsoft Bing, Yahoo, DuckDuckGo, Baidu	**Dogpile** , **Metacrawler** , **Info.com** , **WebCrawler** (historical)

When you search 'NIELIT Web Design Syllabus' on ****Dogpile****: Dogpile submits the query to Google and Bing simultaneously. It finds that both engines return the official NIELIT portal as #1. Dogpile eliminates the duplicate and shows the verified official link at the top.

- A Meta Search Engine does NOT maintain its own database index and does NOT operate web crawlers.
- Meta search engines submit user queries to multiple individual search engines simultaneously.
- Prominent examples of meta search engines include Dogpile, Metacrawler, and Info.com.

QUICK TOPIC RECAP

Meta search engines (e.g. Dogpile, Metacrawler) have no database or crawlers of their own; they query multiple search engines simultaneously and aggregate results.

TOPIC 8.10: WWW ARCHITECTURE & URL ANATOMY

WWW Architecture is the structural framework of protocols, naming systems, and hyperlinked media that powers the Web. A ****Uniform Resource Locator (URL)**** is the standardized global address used to pinpoint the exact location of any digital resource (webpage, image, video, file) on the World Wide Web.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

URLs are the fundamental navigational coordinate system of the Web. Understanding URL anatomy enables developers to structure RESTful routing, pass query parameters, and configure secure HTTPS connections.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

A complete URL consists of 6 standardized segments: 1. Protocol Scheme, 2. Subdomain, 3. Domain Name, 4. Port Number, 5. Directory Path, 6. Resource Filename, and optional 7. Query String / Fragment.

ANATOMY OF A COMPLETE URL BREAKDOWN

URL Segment	Example from: `https://www.nielit.gov.in:443/student/courses.php?id=101#unit3`	Technical Role & Functional Meaning
Protocol / Scheme	`https://`	Hypertext Transfer Protocol Secure; specifies the encrypted communication protocol
Subdomain	`www`	Host machine designation within the parent domain namespace
Domain Name	`nielit.gov.in`	Unique human-readable registered address of the hosting server
Port Number	`:443` (Default for HTTPS; `:80` for HTTP)	Network socket communications port on the target server
Directory Path	`/student/`	Folder hierarchy path on the server file system
Resource File	`courses.php`	Specific file or script being requested and executed
Query String	`?id=101`	Key-value parameters passed to the server script (starts with `?`)

Fragment Identifier	`#unit3`	Bookmark anchor pointing to element with `id="unit3"` on the page
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When you click a link with `?course=m2&batch=2026`: The web server reads `course=m2` from the URL query string, queries the database for the M2-R5.1 batch, and returns customized course notes dynamically.

- URL stands for Uniform Resource Locator.
- Default port for HTTP is 80; default port for HTTPS is 443.
- The query string begins with a question mark `?` and separates parameters with `&`.
- The hash symbol `#` denotes an in-page bookmark fragment identifier.

QUICK TOPIC RECAP

URL anatomy: `Protocol` (https://) + `Domain` (nielit.gov.in) + `Port` (:443) + `Path` (/folder/) + `File` (index.html) + `Query` (?id=1) + `Fragment` (#sec).

TOPIC 8.11: BROWSER ARCHITECTURE AND RENDERING ENGINES

Web Browser Architecture refers to the internal modular software subsystems that coordinate to download, parse, execute, and render web documents onto a computer display. The **Rendering Engine** is the core component responsible for transforming HTML and CSS code into visible pixels.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Understanding browser architecture helps developers write performant code that avoids layout thrashing, minimizes repaints, and works consistently across different browser rendering engines.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Modern web browsers are structured into a 7-layer modular architecture, processing HTML to construct the DOM, CSS to build the CSSOM, combining them into a Render Tree, calculating Layout, and Painting pixels.

THE 7 MODULAR LAYERS OF BROWSER ARCHITECTURE

1. User Interface (UI): The address bar, back/forward buttons, bookmarking menu, refresh button, and developer tools.
2. Browser Engine: Marshals actions and coordinates communication between the User Interface and the Rendering Engine.
3. Rendering Engine: Parses HTML and CSS to render content on screen (e.g. **Blink**, **Gecko**, **WebKit**).
4. Networking: Handles HTTP/HTTPS network requests, SSL certificates, cookies, and DNS socket connections.
5. JavaScript Engine (Interpreter): Parses, compiles (JIT), and executes JavaScript code (e.g. **V8** in Chrome/Edge, **SpiderMonkey** in Firefox, **JavaScriptCore** in Safari).
6. UI Backend: Draws basic interface widgets like select dropdowns and windows using host operating system primitives.
7. Data Persistence: Client-side storage mechanisms including `LocalStorage`, `SessionStorage`, `IndexedDB`, and Cookies.

MAJOR WEB BROWSERS & THEIR RENDERING ENGINES

Web Browser	Primary Rendering Engine	JavaScript Engine	Governing Organization
Google Chrome	Blink (fork of WebKit)	V8 Engine	Google / Chromium Project
Microsoft Edge	Blink (Chromium-based)	V8 Engine	Microsoft
Mozilla Firefox	Gecko	SpiderMonkey	Mozilla Foundation
Apple Safari	WebKit	JavaScriptCore (Nitro)	Apple Inc.
Opera	Blink	V8 Engine	Opera Software

When Chrome renders a page: (1) Networking downloads the HTML bytes. (2) The **Blink Rendering Engine** parses HTML into the DOM tree and CSS into the CSSOM. (3) The **V8 Engine** executes JavaScript scripts. (4) Blink paints the final styled layout on screen.

- The core component that converts HTML and CSS into visible pixels is the Rendering Engine.
- Google Chrome and Microsoft Edge use the Blink rendering engine and V8 JavaScript engine.
- Mozilla Firefox uses the Gecko rendering engine and SpiderMonkey JavaScript engine.
- Apple Safari uses the WebKit rendering engine.

QUICK TOPIC RECAP

Browser architecture: 7 layers including Networking, JS Engine (V8, SpiderMonkey), and Rendering Engine: Blink (Chrome/Edge), Gecko (Firefox), WebKit (Safari).

TOPIC 8.12: HTTP PROTOCOL (METHODS, STATUS CODES, HTTP VS. HTTPS)

HTTP (Hypertext Transfer Protocol) is the foundational application-layer communications protocol of the World Wide Web, operating on a client-server request-response model to exchange web pages, images, and data.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Every browser action relies on HTTP. Understanding HTTP request methods and status codes is essential for debugging web applications, handling form submissions, and securing communications.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

HTTP is **Stateless** (the server retains no memory of past requests). Clients send an HTTP Request containing a Method, URL, and Headers; servers reply with an HTTP Response containing a Status Code and Body.

HTTP VS. HTTPS SECURITY COMPARISON

- HTTP (Port 80): Plain text transmission. Data packets (passwords, credit cards) can be intercepted and read by anyone on the network (packet sniffing).
- HTTPS (Port 443): Hypertext Transfer Protocol Secure. Encrypts all traffic using **SSL / TLS (Transport Layer Security)** certificates, preventing eavesdropping and tampering.
- Visual Indicator: Browsers display a padlock icon in the address bar for valid HTTPS connections.

STANDARD HTTP STATUS CODES REFERENCE

Status Code Range	Category Meaning	Representative Code Examples & Technical Meanings
1xx	Informational	100 Continue (server received initial headers)
2xx	Success	200 OK (request succeeded, page delivered); 201 Created (resource created)
3xx	Redirection	301 Moved Permanently (permanent redirect); 302 Found (temporary redirect); 304 Not Modified
4xx	Client Error	400 Bad Request ; 401 Unauthorized ; 403 Forbidden (access denied); 404 Not Found (file missing)
5xx	Server Error	500 Internal Server Error (crash in PHP/Node code); 502 Bad Gateway; 503 Service Unavailable

When you click a broken link, the server responds with **404 Not Found** (Client error: the file doesn't exist). When a database server crashes, the server responds with **500 Internal Server Error**.

- HTTP is a stateless protocol operating on a client-server model.
- HTTP uses Port 80; HTTPS uses Port 443 with SSL/TLS encryption.
- `200 OK` indicates success; `301` indicates permanent redirect.
- `404 Not Found` is a client error indicating the requested resource does not exist.
- `500 Internal Server Error` indicates server-side code failure.

QUICK TOPIC RECAP

HTTP runs the web (stateless). Methods: GET (fetch), POST (submit). Status codes: 200 (OK), 301 (redirect), 403 (forbidden), 404 (not found), 500 (server crash). HTTPS (port 443) encrypts via SSL/TLS.

TOPIC 8.13: PUBLISHING TOOLS (FILEZILLA, CPANEL, GITHUB)

Web publishing tools are software applications and hosting management interfaces used by web designers to upload, manage, configure, and maintain website files on remote web servers.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Manually copying files to remote servers without specialized tools is complex. Publishing tools provide visual drag-and-drop file transfers, database management, and automated deployments.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Developers connect to servers using FileZilla via FTP/SFTP, manage server databases using cPanel, or deploy code automatically using GitHub Pages.

CONNECTING TO A WEB SERVER WITH FILEZILLA

- Host: The server IP address or domain name (e.g. `ftp.mywebsite.com` or `192.168.1.100`).
- Username: The FTP account username provided by the web hosting service.
- Password: The secret FTP account password.
- Port: `21` for standard unencrypted FTP; `22` for encrypted ****SFTP (SSH File Transfer Protocol)****.
- Quickconnect: Button that establishes the connection and displays server directories.

CORE WEB PUBLISHING TOOLS COMPARED

Publishing Tool	Software Category	Primary Protocol / Mechanism	Practical Web Developer Workflow
FileZilla	Desktop FTP Client (Free Open Source)	FTP (Port 21) & SFTP (Port 22)	Drag-and-drop files from local computer (left panel) to remote server `public_html` (right panel)
cPanel	Web-Based Hosting Control Panel	HTTPS web interface (Port 2083)	Managing databases (phpMyAdmin), email accounts, SSL certificates, DNS, and File Manager directly in browser
GitHub Pages	Git-Based Cloud Hosting Service	Git version control / HTTPS	Free static website hosting directly from a GitHub repository branch; ideal for student portfolios

You finished building an 8-unit O-Level website on your laptop: You open FileZilla, enter your server Host, Username, and Password, click 'Quickconnect', select all files from your desktop folder on the left, and drag them into `public\_html` on the right. Your website is live in 60 seconds!

- FileZilla is a free, cross-platform FTP client used to transfer files to web servers.
- FTP uses Port 21; SFTP (Secure FTP) uses Port 22.
- cPanel is a web-based control panel for managing hosting accounts, databases, and emails.
- GitHub Pages provides free static web hosting directly from Git repositories.

QUICK TOPIC RECAP

Publishing tools: FileZilla (drag-and-drop FTP over port 21 / SFTP port 22), cPanel (web-based server admin and database tools), GitHub Pages (free git-based static hosting).

TOPIC undefined:

- Web Publishing is the 7-phase process of deploying websites: Planning -> Design -> Development -> Testing -> Hosting -> Deployment -> Maintenance.
- Design principles: Simplicity (KISS), Visual Hierarchy, F-shaped scanning pattern, Mobile Responsiveness, and Fast Load Speed (<2 sec).
- SGML (ISO 8879:1986) is the parent metalanguage; HTML is an application of SGML, XML is a lightweight subset of SGML for data interchange.
- Hosting models: Shared (cheapest, shared resources), VPS (virtual isolation), Dedicated (full physical server), Cloud (elastic clustering).
- Interchange standards: 7-bit ASCII (128 chars), universal variable UTF-8 (1-4 bytes for all languages/emojis), MIME headers (`Content-Type`), XML.
- Publishing components: Domain Registrar, DNS name servers, Web Server software (Apache, Nginx, IIS), and Database servers (MySQL).
- Site architecture: Root folder is `public\_html` (or `htdocs`); use relative paths (`images/pic.jpg`); set file permissions to 644 and directories to 755.
- Search engines operate in 3 stages: Crawling (spiders/bots), Indexing (database catalog), and Ranking (algorithms/PageRank). On-page SEO optimizes HTML.
- Meta Search Engines (Dogpile, Metacrawler) have no crawlers or database of their own; they query multiple search engines simultaneously and aggregate results.
- URL anatomy: Protocol (https://) + Domain (nielit.gov.in) + Port (443/80) + Path (/folder/) + File (index.html) + Query (?id=1) + Fragment (#sec).
- Browser architecture has 7 layers including Networking, JS Engine (V8, SpiderMonkey), and Rendering Engine: Blink (Chrome/Edge), Gecko (Firefox), WebKit (Safari).
- HTTP is stateless. Methods: GET (retrieve), POST (submit). Status codes: 200 (OK), 301 (redirect), 403 (forbidden), 404 (not found), 500 (server crash). HTTPS (port 443) encrypts via SSL/TLS.
- Publishing tools: FileZilla (FTP port 21, SFTP port 22), cPanel (web-based hosting control panel), GitHub Pages (git-based free static hosting).

KEY TERMS, ACRONYMS & OFFICIAL FULL FORMS